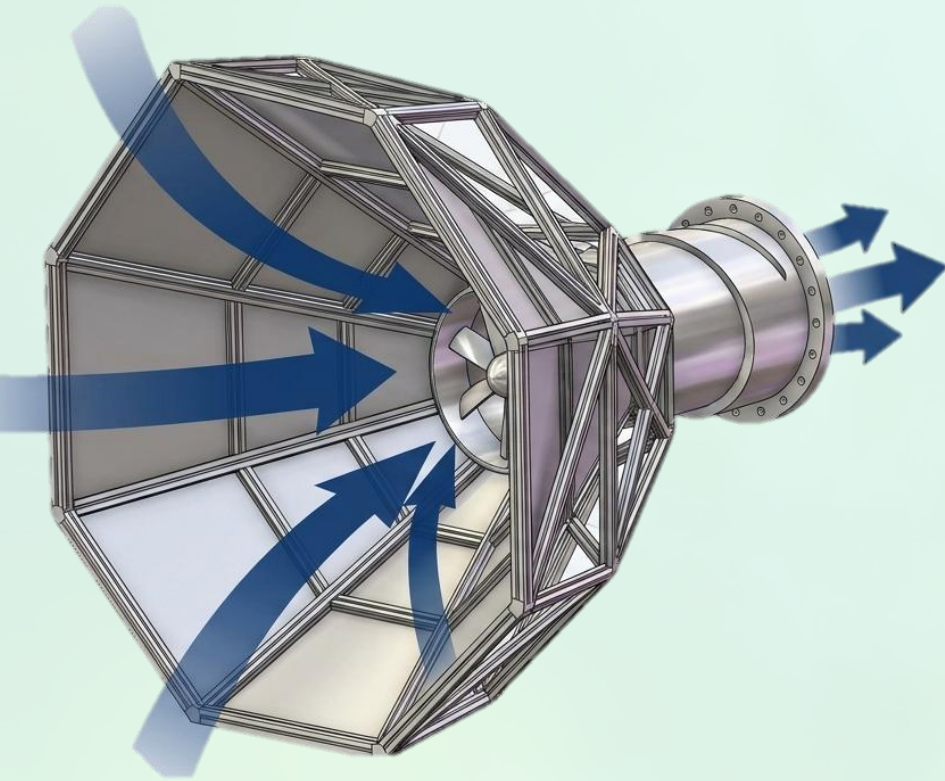




Design & Optimization of a Venturi-Augmented Crossflow Wind Turbine

Group Name : 3110

 ACADEMIC PROJECT

WIJEKOON W.M.T.T.B.

E/21/441

 SIRIMALWATHTHA S.V.D.M.

E/21/383

PREMASINGHE K.B.S.S.

E/21/315

Need & Project Aim



Problem

Conventional turbines require cut in speeds >3.5 m/s and underperform in Sri Lanka's common 2–5 m/s winds.



Existing Tradeoffs

Crossflow rotors are quiet and robust but have lower maximum power coefficients compared to horizontal machines.



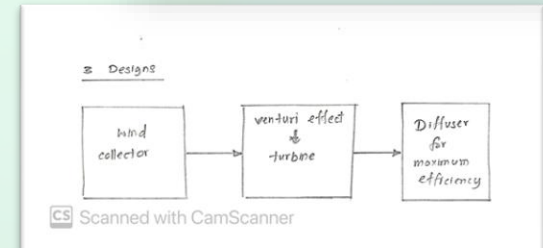
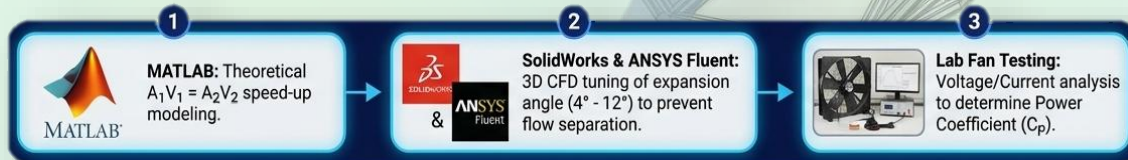
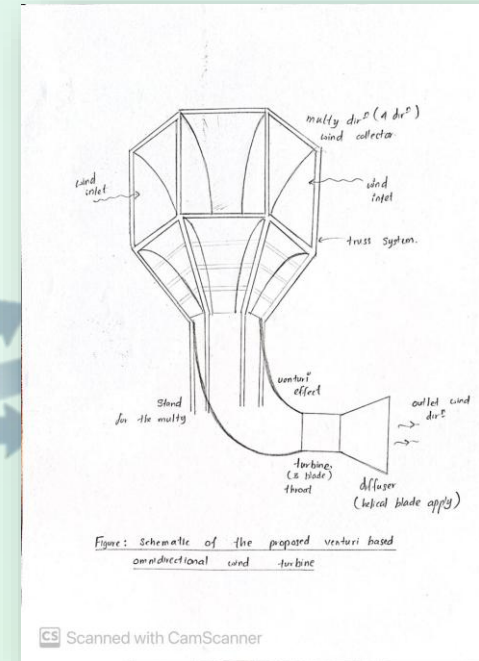
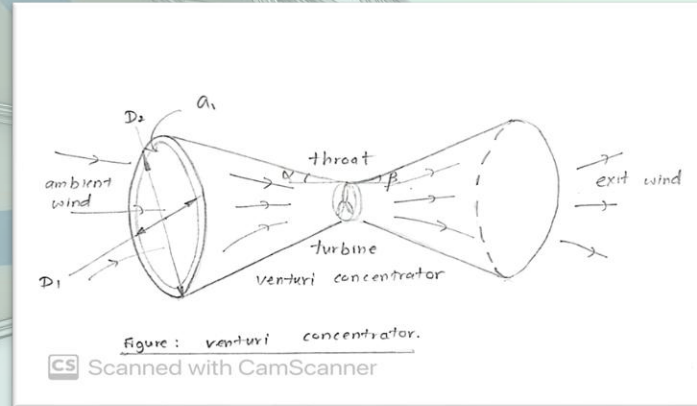
Primary Aim

Create a self starting system using a Venturi style concentrator to boost local wind velocity and energy capture in urban and rural settings.

Engineering Design & Methodology

The design workflow integrates CAD, CFD, and optimization to refine geometry and validate performance:

- SolidWorks: Detailed 3D modeling of a 15-blade rotor
- ANSYS Fluent: CFD studies using the turbulence models
- MATLAB GA: Multi-objective genetic algorithm to optimize blade geometry
- Lab Testing Procedure



Project Timeline

Conceptual Design

Weeks 1-3:
process
Weibull data

Numerical Simulation

Weeks 3-6:
model 15-blade
rotor

Prototype Fabrication

Weeks 6-9:
CFD + GA
optimization

Experimental Testing and Documentation

Weeks 9-14:
validate and
report



Impact Analysis & Next Steps



Economic

Low cost, modular components
(3D printed or fabric frames)
reduce upfront barriers for off
grid communities.



Social

Displaces thermal generation with
Zero emission wind at distributed
scale, supporting national
decarbonization targets.



Environmental

Low aerodynamic noise and reduced
vibration enable deployment in
dense residential areas.